

BoostXL DRV8305 Connector Pinout on LAUNCHXL-F28379D

Table 7. F28379D LaunchPad Pin Out and Pin Mux Options - J5, J7

Mux Value				Mux Value			
X	2	1	0	J5 Pin	J7 Pin	0	X
			3.3V	41	61	5V	
			GPIO95	42	62	GND	
SCIRXDC ⁽¹⁾		NFAULT	GPIO139	43	63	ADCIN15	VSEN A
SCITXDC ⁽¹⁾			GPIO56	44	64	ADCINC5	VSEN B
			GPIO97	45	65	ADCINB5	VSEN C
			GPIO94	46	66	ADCINA5	VSEN PVDD
SPICLKB ⁽¹⁾			GPIO65	47	67	ADCINC4	ISEN A
			GPIO52 ⁽²⁾	48	68	ADCINB4	ISEN B
SCLB ⁽¹⁾			GPIO41 ⁽²⁾	49	69	ADCINA4	ISEN C
SDAB ⁽¹⁾			GPIO40 ⁽²⁾	50	70	ADCINA1	

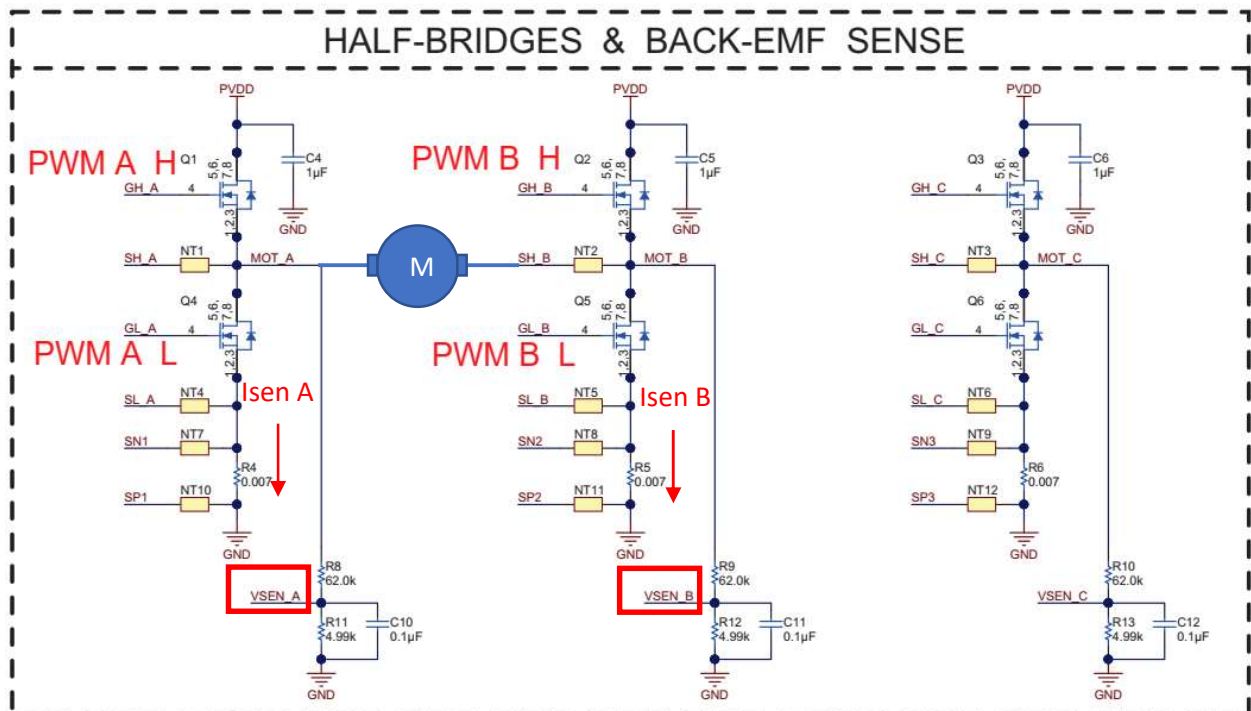
⁽¹⁾ For full pin muxing table for functions shown here and additional mux options, see the [TMS320F2837xD Dual-Core Delfino™ Microcontrollers Data Manual](#).

⁽²⁾ This signal is also routed to the IO expansion header, J9. For information on how to configure this signal, see [Section 5.5](#).

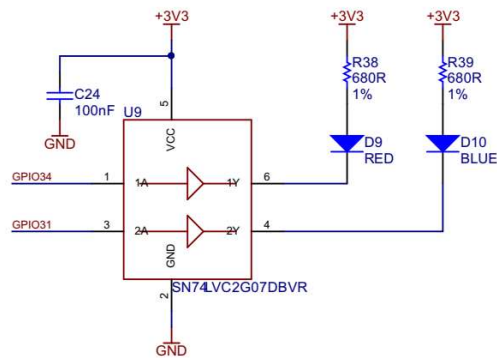
Table 8. F28379D LaunchPad Pin Out and Pin Mux Options - J8, J6

Mux Value				Mux Value			
X	2	1	0	J8 Pin	J6 Pin	0	X
	PWM A H	EPWM4A	GPIO6	80	60	GND	
	PWM A L	EPWM4B	GPIO7	79	59	GPIO66	
	PWM B H	EPWM5A	GPIO8	78	58	GPIO131	SD2_C1 ⁽¹⁾
	PWM B L	EPWM5B	GPIO9	77	57	GPIO130	SD2_D1 ⁽¹⁾
		EPWM6A	GPIO10	76	56	RST	
		EPWM6B	GPIO11	75	55	GPIO63	SPISIMOB ⁽¹⁾
OUTPUTXBAR3 ⁽¹⁾			GPIO14	74	54	GPIO64	SPISOMIB ⁽¹⁾
OUTPUTXBAR4 ⁽¹⁾			GPIO15	73	53	GPIO26	SD2_D2 ⁽¹⁾
			DAC3	72	52	GPIO27	SD2_C2 ⁽¹⁾
			DAC4	71	51	GPIO25	OUTPUTXBAR2 ⁽¹⁾

H-Bridge



LED

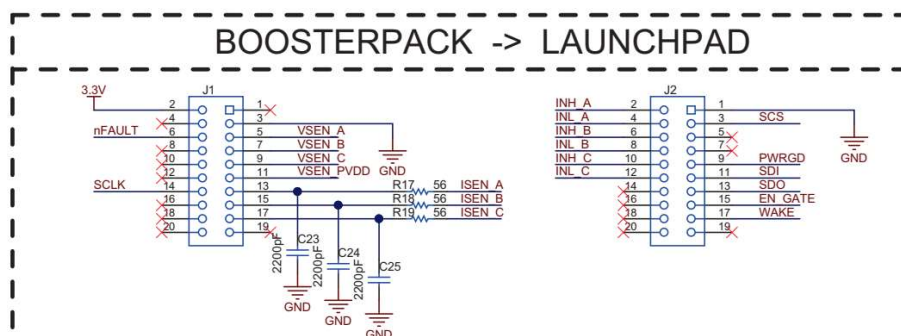


Other Signal and pinout

5.3 BoosterPack GPIO Signals

The Motor Drive BoosterPack brings out the GPIO signals for the DRV8301 to the LaunchPad XL. These signals are described in detail in the following list and further information can be found in the DRV8301 data sheet.

1. **nFAULT**: Fault indicator, specific FAULT status can be obtained through the status registers
2. **PWRGD**: Watchdog and LDO regulator status indicator
3. **EN_GATE**: Enables gate driver and current shunt amplifiers
4. **WAKE**: Used to bring the device out of its low power sleep mode.



2.2 Pinout

The BOOSTXL-DRV8305EVM brings out a mixture of power, control, and feedback signals to the XL LaunchPad headers.

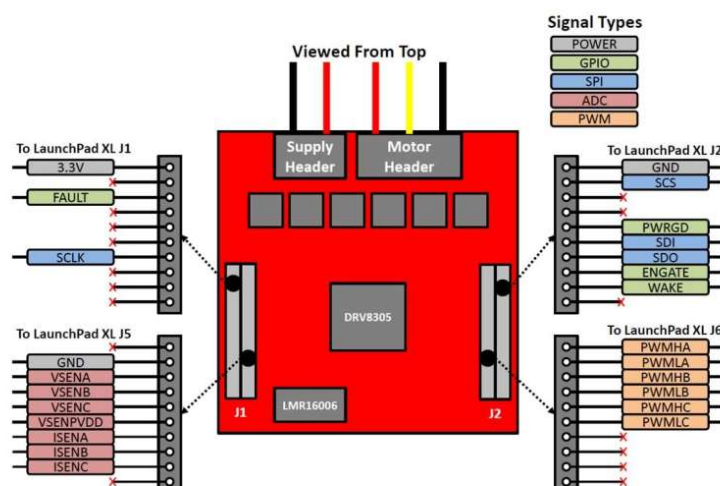


Figure 6. BOOSTXL-DRV8305EVM Pinout

5.2 Low-Side Current Shunt Sense

The BoosterPack has low-side current shunt sense for each half-bridge (phases A, B, and C). The current sense setup takes advantage of the DRV8305's triple shunt current amplifiers (phases A, B, and c). The configuration for the low-side sense is shown in Figure 9. The differential amplifier senses voltage across a $0.007\text{-}\Omega$ power sense resistor with differential connections. The differential voltage is then amplified by 10 V/V and centered at 1.65 V to allow for sensing both positive and negative currents. The sense resistor has been scaled for 0- to 20-A peak currents.

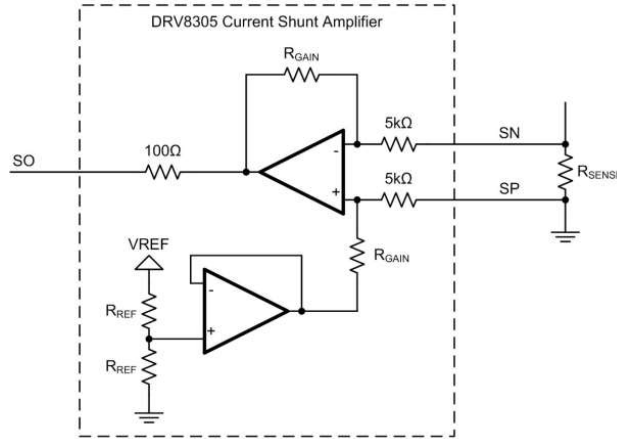


Figure 10. Current Sense

5.1 DC Bus and Phase Voltage Sense

The BoosterPack has been designed with voltage sense circuits on the DC bus (PVDD) and each half-bridge outputs (phases A, B, and C). These circuits, shown Figure 9, consist of a voltage divider with a filtering capacitor to reduce high frequency noise on the ADC pins. These circuits have been scaled to support 4.4 to 45 V . The high-side resistors for the phase outputs are located near the motor output header (J4) while the low-side resistors and filtering capacitors are located near the ADC pins on the BoosterPack to LaunchPad header (J1) for improved noise reduction purposes.

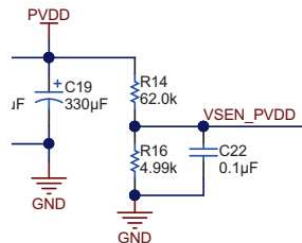


Figure 9. Voltage Sense